

	Since F and a are constants, F varies linearly with t .
9	D Both boats travel the same distance s and experience the same force F. Hence the total work done by the force F between the starting line and the finishing line is the same for each boat – they will have the same final kinetic energy. Boat with mass m will experience a larger acceleration, and will thus reach the finish line first. Note: Boat with mass m will also have a higher speed, since the two boats have the same final kinetic energy.

10	B
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